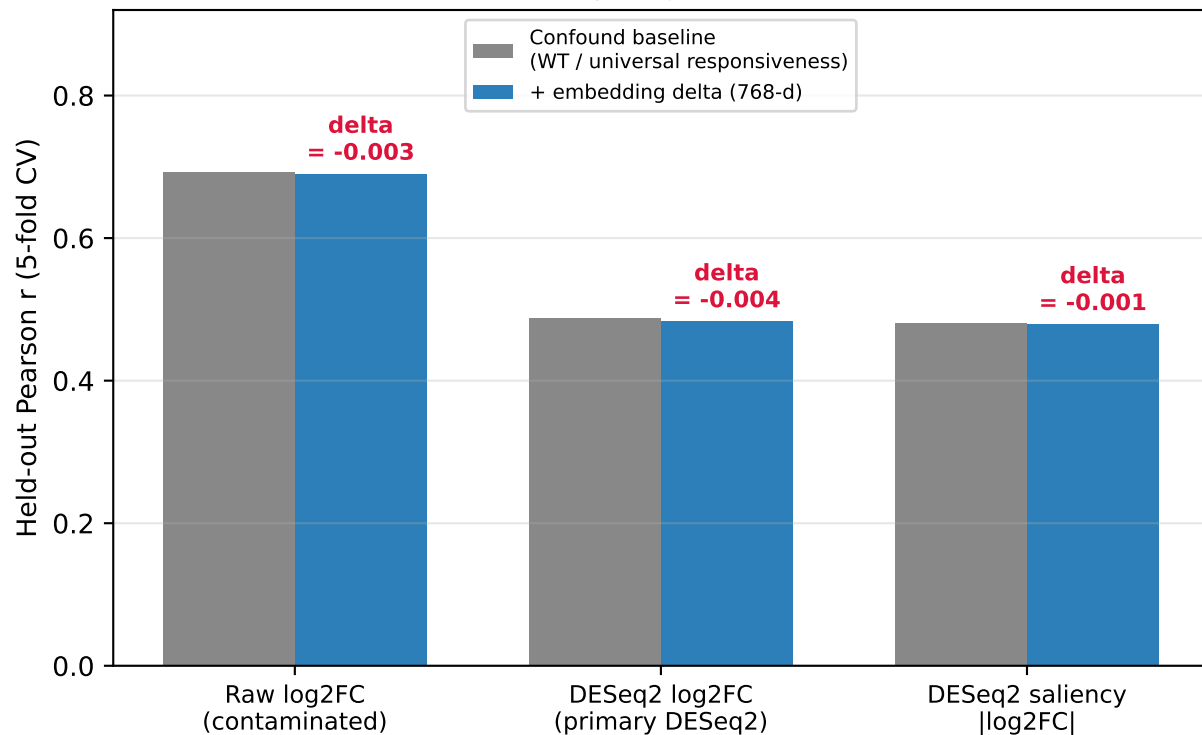


Frangieh: Geneformer in-silico KO embedding delta carries no perturbation-specific signal

A. Embedding delta adds no measurable increment beyond confounds (across every target and readout)



B. Every apparent positive signal collapses under control (controlled estimates approach zero across readouts)

